

Homework 3: Probability, and testing independence

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Simulation

All simulations use numpy's default_rng with a fixed seed of 42, so every number below is reproducible by running HW3.py.

1. Two dice: P(4-sided | rolled a 1)

seed 42, 100,000 trials, 20,633 rolls of 1
simulated P(4-sided | rolled 1) = 0.5953
exact = 0.6000
difference = -0.0047

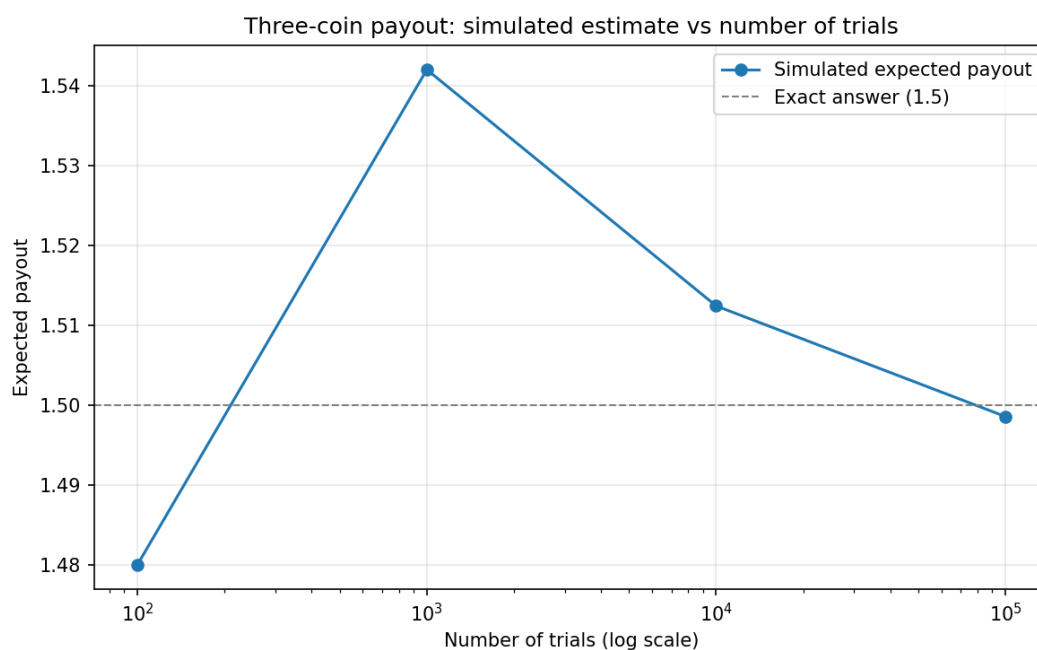
The exact answer is $(1/2 \times 1/4) / (1/2 \times 1/4 + 1/2 \times 1/6) = 0.6$. The estimate rests on the 20,633 trials that rolled a 1, not all 100,000, so its standard error is about 0.003 and a miss of 0.005 is ordinary noise.

2. Three-coin payout

simulated expected payout = 1.5030
exact = 1.5000
difference = +0.0030

3. Estimate vs number of trials

trials	estimate
100	1.4800
1,000	1.5420
10,000	1.5124
100,000	1.4985



At 100 trials the estimate is off by 0.02, and at 1,000 it is off by 0.04; it only settles within a few thousandths of 1.5 at 100,000. The error shrinks roughly with the square root of the number of trials, so each extra digit of accuracy costs 100 times more data. A backtest over a few hundred days carries the same kind of noise.

Are daily returns independent?

4. Ten years of SPY

first date: 2016-09-30, last date: 2026-09-28
trading days (returns): 2511
mean daily return: 0.0634%

Daily adjusted closes from yfinance. A day is "down" if its return is below zero. Each (today, tomorrow) pair of consecutive trading days is counted by hand in a loop, which gives 2,510 pairs from 2,511 returns.

5. P(tomorrow down) and P(tomorrow down | today down)

P(tomorrow down) = 0.4454 (1118 down days out of 2510)
P(tomorrow down | today down) = 0.4387 (490 out of 1117 down days)

6. P(tomorrow down | today down more than 2%)

P(tomorrow down | today down > 2%) = 0.3908 (34 out of 87 days)

Only 87 of 2,510 days, about 3.5%, were drops of more than 2%. That is too few to say much: with 87 days the standard error on this probability is about 0.05, so 0.39 could really be anywhere from about 0.29 to 0.50.

Expected present value

7. expected_present_value

expected_present_value([10, 10, 10], 0.10, 0.5) = 15.10

Year t 's cash flow is weighted by $\text{survival_prob}^{t-1}$, because the company has to survive every year before it: $10/1.1 + 0.5 \times 10/1.1^2 + 0.25 \times 10/1.1^3 = 9.09 + 4.13 + 1.88 = 15.10$. Applying the 0.5 once to both later years gives the Session 3 figure of 16.98 instead. With $\text{survival_prob} = 1$ the function returns exactly `present_value` from Homework 2 (checked with an `assert`).

Written

8. Are daily returns independent?

The direction of SPY's returns looks close to independent, but returns as a whole are not. After a down day, the chance of another down day is 43.9% (1,117 days), almost the same as the 44.5% base rate. The 0.7-point gap is less than half a standard error, which is well within noise. After a drop of more than 2%, the chance of another down day falls to 39.1%. That hints at a rebound, but it rests on only 87 days, so it is about one standard error from the base rate and not a result I would trust. The size of moves is clearly not independent, though. Using the same counting, a move of more than 2% in either direction happens on 6.0% of days (150 of 2,510). The day after a drop of more than 2%, it happens 27.6% of the time (24 of 87), more than four times as often. Volatility clusters: calm days tend to follow calm days, and violent days follow violent days. So a strategy that assumes independence may be fine on direction, since yesterday's sign tells you almost nothing about today's. But it will be badly wrong on risk. If it sizes positions or sets stops

using one average volatility, it takes too much risk right after a shock, just when the next big move is most likely. It will also underestimate how often losses bunch together into drawdowns. A strategy that bets on the post-crash rebound is relying on 87 days of evidence, so it is much weaker than it looks.